

# Exercise 1 — Kotlin Jetpack Compose Mobile Shop

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Course: COMP5450 Mobile Programming

Tech stack: Kotlin + Jetpack Compose

## Assignment summary

This project implements a mobile shop app in Kotlin Jetpack Compose. The app uses in-memory sample data, so no database is required. Users can browse shoes, open a product detail screen, choose color and size, add items to the bag, and see the total price in the shopping bag tab.

## Implemented features

- Splash / landing screen inspired by the provided mockup.
- Home screen with featured banner, category chips, and product grid.
- Product detail screen with color selector, size selector, rating, and add-to-bag button.
- Shopping bag tab with selected items, remove action, and total price calculation.
- In-memory catalog for simple course-friendly data management.

## How to run

- 1 Open `/Users/lmc/repos/mobile_programming/exercises/exercise-1` in Android Studio.
- 2 Use JDK 17 for Gradle.
- 3 Allow Gradle sync to complete.
- 4 Run the app on an Android emulator or physical device.
- 5 You can also build from the terminal with `./gradlew :app:assembleDebug` after setting `JAVA_HOME` to JDK 17.

## Project structure

`MainActivity.kt` starts the app. The `data` package contains product and cart models plus sample data. The `ui` package contains the `ViewModel`, navigation host, theme files, reusable product artwork, and the Compose screens.

## Build output

A debug APK was successfully built at:

`/Users/lmc/repos/mobile_programming/exercises/exercise-1/app/build/outputs/apk/debug/app-debug.apk`

## Submission notes

- Upload the full Android Studio project to a public GitHub repository.
- Upload a ZIP to D2L containing the project files required by the instructor.

- Include screenshots from your emulator or real device before final submission if the instructor requires them in the PDF package.